Section D – AI-Augmented Learning

Task: AI-Assisted Subnetting Plan

1. Exact Prompt Used

"Create a subnetting plan for a small office network containing 5 computers and 1 network printer. Use a Class C private IP network. Provide the subnet address, subnet mask, usable IP address range, broadcast address, and assign a unique IP address to each computer and printer. Keep some IP addresses reserved for future expansion."

2. Original Subnetting Plan Generated by AI

Network Details:

- Network Address: 192.168.1.0
- Subnet Mask: 255.255.255.0
- CIDR Notation: /24
- Usable IP Range: 192.168.1.1 – 192.168.1.254
- Broadcast Address: 192.168.1.255
- Total Usable IP Addresses: 254

IP Address Allocation:

Device	IP Address
Computer 1	192.168.1.10
Computer 2	192.168.1.11
Computer 3	192.168.1.12
Computer 4	192.168.1.13
Computer 5	192.168.1.14
Network Printer	192.168.1.20

3. Independent Verification and Corrected Plan

The AI-generated subnetting plan was checked independently.

- The network address and broadcast address are correct.
- All assigned IP addresses are within the usable host range.

- No IP address is duplicated.
- The subnet supports 254 usable host addresses, which is sufficient for the current devices and future expansion.

Corrected Subnetting Plan:

No corrections were required because the original AI-generated plan was technically correct.

Device	Assigned IP Address
Computer 1	192.168.1.10
Computer 2	192.168.1.11
Computer 3	192.168.1.12
Computer 4	192.168.1.13
Computer 5	192.168.1.14
Network Printer	192.168.1.20

Reserved Addresses for Future Expansion: 192.168.1.21 – 192.168.1.50

4. Explanation

AI helped generate the initial subnetting plan and allocate IP addresses to all network devices. I independently verified the subnet mask, usable IP range, broadcast address, and device allocation. No errors or overlapping IP addresses were found. The plan provides enough addresses for the existing devices and future network expansion.